PDF Duplex Merge Tool - PLAN.md

Project Goal

Create a Python utility that merges two PDF files created from a simplex scanner workflow into a correctly ordered duplex PDF.

Typical Workflow

Original document:

Sheet 1: Front=Page 1 Back=Page 2 Sheet 2: Front=Page 3 Back=Page 4 Sheet 3: Front=Page 5 Back=Page 6

Scan Operation

Pass 1

Scan all fronts:

front.pdf Page 1 Page 3 Page 5

Pass 2

Flip stack and scan backs:
back.pdf Page 6 Page 4 Page 2

Desired Result

merged.pdf Page 1 Page 2 Page 3 Page 4 Page 5 Page 6

Phase 1 - Core Merge Engine

Inputs

front.pdf back.pdf

Back Scan:
4 2

The program must detect:
front count = back count + 1

and merge correctly.

Result:

1 2 3 4 5

Phase 3 - Validation

Detect:

Missing Pages

Example:

front = 20 pages back = 15 pages

Warning:
Page counts do not match. Possible scanning error.

Empty PDFs

0 pages

Generate clear error.

Encrypted PDFs

Detect and report.

- Title
- Author
- Subject

to output PDF.

Phase 7 - GUI (Future)

Simple desktop app.

```
+---------------------------+
| Front PDF  [Browse]       |
| Back PDF   [Browse]       |
| Output PDF [Browse]       |
|                           |
| [ Merge ]                 |
+---------------------------+
```

Possible toolkits:

- Tkinter
- PySide6
- PyQt6

Technical Stack

Language

Python 3.12+

PDF Library

Preferred: pypdf

Reason:

- Active project
- Pure Python
- Good PDF support

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